

COSC 594

DM Alignment (II)

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Outline

- RECAP
- PREFERENCE EXAMPLES
- PREFERENCE-BASED ALIGNMENT

Outline

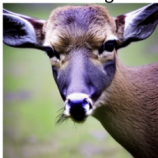
■ RECAP

Downstream requirements

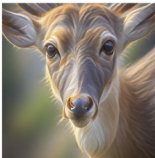
■ AESTHETIC QUALITY

Prompt: *A photo of a deer*

No fine-tuning

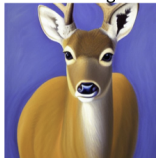


Aesthetic reward

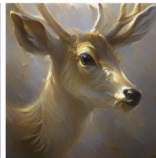


Prompt: *A painting of a deer*

No fine-tuning



Aesthetic reward



■ HUMAN PREFERENCE



a painting of an old man



a painting of a racoon
washing dishes

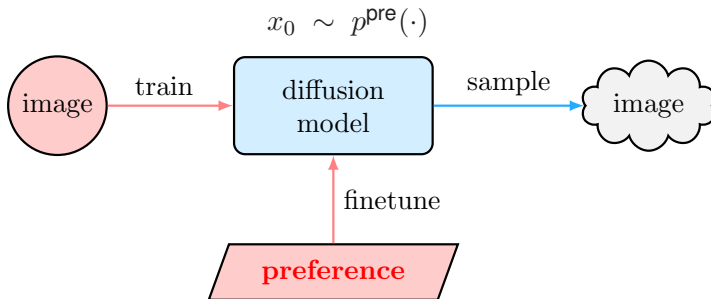
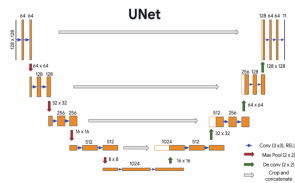


a painting of a tree



a painting of a city skyline

DM alignment framework



preference – reward / human value

Reward-based alignment

Step 1: Diffusion Model Pretraining



Step 2: Reward Modeling



Step 3: Reinforcement Learning-Based Fine-Tuning

Fine-tuning w/ path-wise KL-regularization

Trajectory distribution: $p(x_{0:T}) = \prod_{t=1}^T p_t(x_{t-1} | x_t) p_{T+1}(x_T)$

Target distribution: $p(x_0) = \int_{x_{1:T}} p(x_{0:T}) d_{x_{1:T}}$

Reward model: $r(x_0)$ for a sample x_0

■ DISTRIBUTION OPTIMIZATION

$$\underset{p}{\text{maximize}} \quad \mathbb{E}_{x_0 \sim p(\cdot)} [r(x_0)] - \beta D_{\text{KL}}(p(x_{0:T}) \parallel p^{\text{pre}}(x_{0:T}))$$

$\beta > 0$ – tradeoff parameter

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High quality images

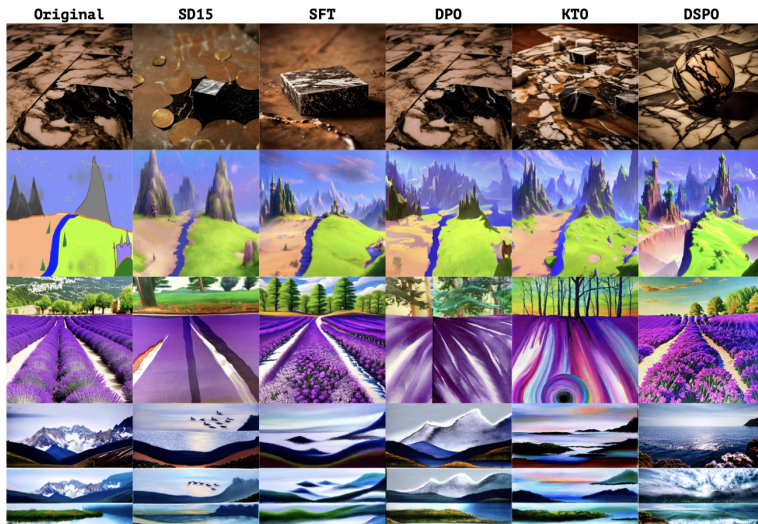


image source: <https://openreview.net/forum?id=xyfb9HHvMe> (DSPO)

High quality details

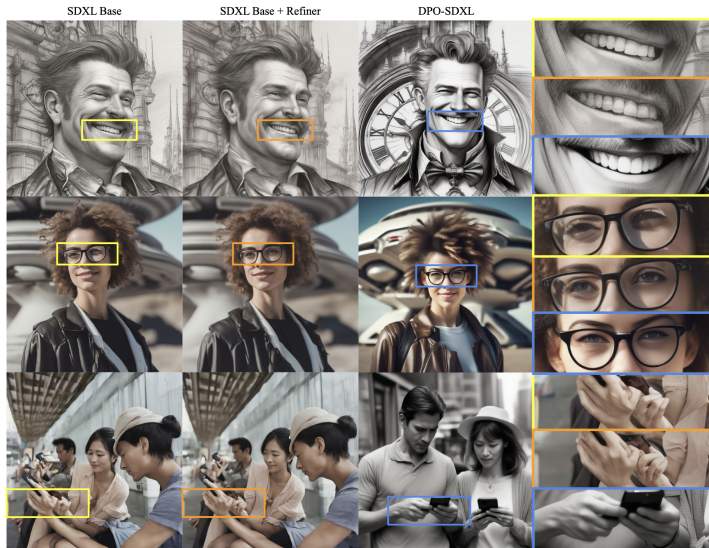
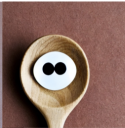
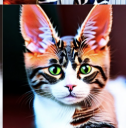


image source: arXiv:2311.12908 (DPO)

Text alignment

Prompt	SD1.5	SFT	Diff-DPO	Diff-KTO	DSP0	DMPO
<i>A spoon dressed up with eyes and a smile.</i>						
<i>A papaya fruit dressed as a sailor.</i>						
<i>photo of a zebra dressed suit and tie sitting at a table in a bar with a bar stools, award winning photography.</i>						
<i>A cat with two horns on its head.</i>						

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Path-wise preference

Pretrained diffusion model: $p_t^{\text{pre}}(x_{t-1} \mid x_t)$ for $t = T, \dots, 2, 1$

Trajectory or path: $x_T \rightarrow x_{T-1} \rightarrow \dots \rightarrow x_1 \rightarrow x_0$

Preference: $x_{0:T}^w \succ x_{0:T}^l$

■ BRADLEY-TERRY MODEL

$$\mathbb{P}(x_{0:T}^w \succ x_{0:T}^l) = \frac{e^{r(x_{0:T}^w)}}{e^{r(x_{0:T}^w)} + e^{r(x_{0:T}^l)}} := \sigma(r(x_{0:T}^w) - r(x_{0:T}^l))$$

$$\sigma(t) = 1/(1 + e^{-t})$$

Consistency between reward and preference?

Path-wise reward training

Data set : $D = \{x_{0:T}^{(i,w)}, x_{0:T}^{(i,l)}\}_{i=1}^N$

Reward model: $r_\phi(x_{0:T})$ w/ parameter ϕ

■ LIKELIHOOD MAXIMIZATION

$$\underset{\phi}{\text{maximize}} \quad \mathcal{L}(r_\phi; D) \quad := \quad \mathbb{E}_{(x_{0:T}^w, x_{0:T}^l) \sim D} \left[\log \mathbb{P}(x_{0:T}^w \succ x_{0:T}^l) \right]$$

Perfect reward represents preference

Fine-tuning w/ path-wise KL-regularization

■ DISTRIBUTION OPTIMIZATION

$$\underset{p}{\text{maximize}} \quad \mathbb{E}_{x_0 \sim p(\cdot)}[r(x_0)] - \beta D_{\text{KL}}(p(x_{0:T}) \parallel p^{\text{pre}}(x_{0:T}))$$

$r(x_{0:T}) = r(x_0)$ – path-wise reward

$\beta > 0$ – tradeoff parameter

★ optimal model p^*

$$p^*(x_{0:T}) = p^{\text{pre}}(x_{0:T}) e^{r(x_{0:T})/\beta} \frac{1}{Z}$$

$Z := \int_{x_{0:T}} p^{\text{pre}}(x_{0:T}) e^{r(x_{0:T})/\beta} dx_{0:T}$ – normalization constant

Intractable **normalization**

Correspondence between model and reward

$$p^{\star}(x_{0:T}) = p^{\text{pre}}(x_{0:T}) e^{r(x_{0:T})/\beta} \frac{1}{Z}$$

- ★ from reward to model
- ★ from model to reward

$$r(x_{0:T}) = \beta \log \frac{p^{\star}(x_{0:T})}{p^{\text{pre}}(x_{0:T})} + \beta \log Z$$

Correspondence, up to a constant

Reward representation trick

Data set : $\tilde{D} = \{x_{0:T}^{(i,w)}, x_{0:T}^{(l,i)}\}_{i=1}^N$

Reward modeling: $\text{maximize}_r \mathbb{E}_{(x_{0:T}^w, x_{0:T}^l) \sim \tilde{D}} [\log \mathbb{P}(x_{0:T}^w \succ x_{0:T}^l)]$

■ REWARD REPRESENTATION

$$\begin{aligned} & \mathbb{E}_{(x_{0:T}^w, x_{0:T}^l) \sim \tilde{D}} [\log \mathbb{P}(x_{0:T}^w \succ x_{0:T}^l \mid x)] \\ &= \mathbb{E}_{(x_{0:T}^w, x_{0:T}^l) \sim \tilde{D}} [\log \sigma(r(x_{0:T}^w) - r(x_{0:T}^l))] \\ &= \mathbb{E}_{(x_{0:T}^w, x_{0:T}^l) \sim \tilde{D}} \left[\log \sigma \left(\beta \log \frac{p^*(x_{0:T}^w)}{p^{\text{pre}}(x_{0:T}^w)} - \beta \log \frac{p^*(x_{0:T}^l)}{p^{\text{pre}}(x_{0:T}^l)} \right) \right] \end{aligned}$$

★ preference-based objective

$$\text{maximize}_p \mathbb{E}_{(x_{0:T}^w, x_{0:T}^l) \sim \tilde{D}} \left[\log \sigma \left(\beta \log \frac{p(x_{0:T}^w)}{p^{\text{pre}}(x_{0:T}^w)} - \beta \log \frac{p(x_{0:T}^l)}{p^{\text{pre}}(x_{0:T}^l)} \right) \right]$$

Fine-tuning w/ path-wise KL-regularization in parameter space

■ PARAMETER OPTIMIZATION

$$\underset{\theta}{\text{maximize}} \quad \mathbb{E}_{(x_{0:T}^w, x_{0:T}^l) \sim \tilde{D}} \left[\log \sigma \left(\beta \log \frac{p_{\theta}(x_{0:T}^w)}{p^{\text{pre}}(x_{0:T}^w)} - \beta \log \frac{p_{\theta}(x_{0:T}^l)}{p^{\text{pre}}(x_{0:T}^l)} \right) \right]$$

★ nonconvex parameter optimization

$$\hat{r}(x_{0:T}) := \beta \log \frac{p_{\theta}(x_{0:T})}{p^{\text{pre}}(x_{0:T})} \quad - \text{reward parametrization}$$

■ POLICY GRADIENT (DIRECT PREFERENCE OPTIMIZATION, DPO)

$$\begin{aligned} & \nabla_{\theta} \mathbb{E}_{(x_{0:T}^w, x_{0:T}^l) \sim \tilde{D}} \left[\log \sigma \left(\beta \log \frac{p_{\theta}(x_{0:T}^w)}{p^{\text{pre}}(x_{0:T}^w)} - \beta \log \frac{p_{\theta}(x_{0:T}^l)}{p^{\text{pre}}(x_{0:T}^l)} \right) \right] \\ &= \mathbb{E}_{(x_{0:T}^w, x_{0:T}^l) \sim \tilde{D}} \left[\sigma \left(\hat{r}(x_{0:T}^w) - \hat{r}(x_{0:T}^l) \right) \left(\nabla_{\theta} \log p_{\theta}(x_{0:T}^w) - \nabla_{\theta} \log p_{\theta}(x_{0:T}^l) \right) \right] \end{aligned}$$

Outline

■ RECAP

■ PREFERENCE EXAMPLES

High quality images/details, text alignment

■ PREFERENCE-BASED ALIGNMENT

Path-wise preference, path-wise DPO